

BG77xA-GL&BG95xA-GL

QCFGEXT AT Commands Manual

LPWA Module Series

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About the Document

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1 Introduction

This document describes the **AT+QCFGEXT** commands supported on BG770A-GL, BG772A-GL, BG950A-GL, BG951A-GL and BG952A-GL modules.

1.1. Special Marks

Table 1: Special Mark

Mark	Definition
*	Unless otherwise specified, when an asterisk (*) is used after a function, feature, interface, pin name, AT command, or argument, it indicates that the function, feature, interface, pin, AT command, or argument is under development and currently not supported; and the asterisk (*) after a model indicates that the sample of such a model is currently unavailable.

1.2. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

1.3. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this

document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

AT+QCFGEXT implemented by BG770A-GL, BG772A-GL, BG950A-GL, BG951A-GL and BG952A-GL modules fall in “Extended” category syntactically, as illustrated below.

- **Extended**

There are several types of extended commands as shown in the following table:

Table 2: Types of AT Commands and Responses

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

Multiple commands can be placed on a single line using a semi-colon (;) between commands. In such cases, only the first command should have **AT** prefix. Commands can be in upper or lower case.

Spaces should be ignored when you enter AT commands, except in the following cases:

- Within quoted strings, where spaces are preserved;
- Within an unquoted string or numeric parameter;
- Within an IP address;
- Within the AT command name up to and including =, ? or =?.

On input, at least a carriage return is required. A newline character is ignored so it is permissible to use carriage return/line feed pairs on the input.

If no command is entered after the **AT** token, **OK** will be returned. If an invalid command is entered, **ERROR** will be returned.

Optional parameters, unless explicitly stated, need to be provided up to the last entered parameter.

1.4. AT Command Responses

When the AT command processor has finished processing a line, it will output **OK**, **ERROR** or **+CME ERROR: <err>** to indicate that it is ready to accept a new command. Solicited information responses are sent before the final **OK**, **ERROR** or **+CME ERROR: <err>**.

Response format:

```
<CR><LF>+CMD1:<parameters><CR><LF>  
<CR><LF>OK<CR><LF>
```

Or

```
<CR><LF><parameters><CR><LF>  
<CR><LF>OK<CR><LF>
```

1.5. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2 Description of AT+QCFGEXT Commands

2.1. AT+QCFGEXT Extended Configuration Settings

The following Test Command shows the supported extended configuration settings of the module.

AT+QCFGEXT Extended Configuration Settings	
Test Command	Response
AT+QCFGEXT=?	+QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] +QCFGEXT: "copstout",(range of supported <timeout>s) +QCFGEXT: "deletegeo",<geoid> +QCFGEXT: "querygeo",<geoid> +QCFGEXT: "nipdcfg",<type>,<apn>,<username>,<password>]]] +QCFGEXT: "nipd",<mode>,<timeout>]] +QCFGEXT: "nipds",<mode>,<data>,<data_length>]] +QCFGEXT: "nipdr",<read_length>,<read_mode>]] +QCFGEXT: "quecopen",<list of supported <value>s]) +QCFGEXT: "fota_apn",<iptype>,<apn>,<username>,<password>] +QCFGEXT: "stats",<layer>,<filter>,<counter_type>,<action>]]] +QCFGEXT: "ver" +QCFGEXT: "debug",<range of supported <value>s]) +QCFGEXT: "devinfo" +QCFGEXT: "meas",<list of supported <measurement_type>s]) +QCFGEXT: "setbdelay",<list of supported <mode>s]) +QCFGEXT: "xonoff",<list of supported <sw_flow_en>s]) OK

2.1.1. PS-Related AT Commands

2.1.1.1. AT+QCFGEXT="nipdcfg" Configure NIDD Connection

This command configures an NIDD connection or queries the current setting.

AT+QCFGEXT="nipdcfg" Configure NIDD Connection	
Write Command AT+QCFGEXT="nipdcfg"[,<type>[,<apn>[,<username>,<password>]]]	Response If the optional parameters are omitted, query the current setting: +QCFGEXT: "nipdcfg",<type>,<apn> OK If any of the optional parameters is specified, configure the NIDD connection: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<type>	Integer type. Non-IP outgoing data type. 0 MO Non-IP data type 1 MO Exception Non-IP data type
<apn>	String type. Access point name.
<username>	String type. Username of the selected APN.
<password>	String type. Password of the selected APN.

2.1.1.2. AT+QCFGEXT="nipd" Open/Close NIDD Connection

This command opens or closes an NIDD connection.

AT+QCFGEXT="nipd" Open/Close NIDD Connection	
Write Command AT+QCFGEXT="nipd",<mode>[,<tim	Response If <mode>=0, close the NIDD connection:

eout>]	OK If <mode>=1, open the NIDD connection: OK +QIND: "nipd","open",<errcode> If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<mode>	Integer type. Close or open an NIDD connection. 0 Close the NIDD connection 1 Open the NIDD connection
<timeout>	Integer type. Timeout for opening the NIDD connection. Parameter is valid only when <mode>=1. Range: 30–90. Default: 30. Unit: second.
<errcode>	Integer type. Result code. See Chapter 3 for more information.

2.1.1.3. AT+QCFGEXT="nipds" Send MO Non-IP Data

This command sends MO Non-IP data to a server.

AT+QCFGEXT="nipds" Send MO Non-IP Data	
Write Command AT+QCFGEXT="nipds",<mode>,<data>[,<data_length>[,<rai>]	Response OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms

Characteristics

The command takes effect immediately.
The configurations are not saved.

Parameter

<mode>	Integer type. <data> format. 0 ASCII format string 1 Hex format string
<data>	ASCII format string or hex format string. Data to be sent.
<data_length>	Integer type. Length of the data to be sent. Range: 1-1358 for ASCII format; 1-679 for hex format. Unit: byte. If this parameter is omitted, <data> can be specified at any length within 1358 bytes of ASCII format, or within 679 bytes of hex format.
<errcode>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.4. AT+QCFGEXT="nipdr" Retrieve MT Non-IP Data

This command retrieves the data reported by the URC **+QIND: "nipd","recv"**.

AT+QCFGEXT="nipdr" Retrieve MT Non-IP Data

Write Command

**AT+QCFGEXT="nipdr",<read_length>
>[,<read_mode>]]**

Response

+QCFGEXT: "nipdr",<read_actual_length>,<data>

OK

If there are no data that can be retrieved:

+QCFGEXT: "nipdr",0

OK

If there is an error related to ME functionality:

+CME ERROR: <result>

If there is any other error:

ERROR

Write Command

When **<read_length>** is 0, query the read status of the retrieved data:

AT+QCFGEXT="nipdr",0

Response

If the connection has existed:

+QCFGEXT: "nipdr",<total_receive_length>,<have_read_length>,<unread_length>

OK

	If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<read_length>	Integer type. The length of the data to be retrieved. Retrieve all available data if this parameter is omitted. Unit: byte.
<read_mode>	Integer type. <data> format. This parameter is valid only when <read_length> is not 0. 0 String type 1 Hex type
<read_actual_length>	Integer type. Actual length of retrieved data. Unit: byte.
<data>	String type. Retrieved data in format chosen with <read_mode>.
<total_receive_length>	Integer type. Total length of received data. Unit: byte.
<have_read_length>	Integer type. Length of retrieved data. Unit: byte.
<unread_length>	Integer type. Length of unread data. Unit: byte.
<errcode>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.5. AT+QCFGEXT="fota_apn" Configure IP Family and APN for DFOTA

This command configures the IP family and APN for DFOTA or queries the current setting.

AT+QCFGEXT="fota_apn" Configure IP Family and APN for DFOTA

Write Command	Response
AT+QCFGEXT="fota_apn"[,<iptype>,<apn>,<username>,<password>]	If the optional parameters are omitted, query the current setting: +QCFGEXT: "fota_apn",<iptype>,<apn>,<username>,<password> OK If any of the optional parameters is specified, set the IP family and APN for DFOTA: OK If there is any error:

	ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<iptype>	Integer type. IP family. 0 IPv4 address family 1 IPv6 address family 2 IPv4 and IPv6 address family
<apn>	String type. Access point name.
<username>	String type. Username of the selected APN.
<password>	String type. Password of the selected APN.

2.1.1.6. AT+QCFGEXT="stats" Return Counters per LTE Protocol Layer/Sub-system

The command returns counters per LTE protocol layer/sub-system or clears the counters.

AT+QCFGEXT="stats" Return Counters per LTE Protocol Layer/Sub-system	
Write Command AT+QCFGEXT="stats",<layer>[, [<filter>], [<counter_type>], [<action>]]]	Response If <action> is omitted, return counters per LTE protocol layer/sub-system: +QCFGEXT: "stats",<stats_string> OK If <action>="CLEAR" , clear the counters: OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<layer>	String type. LTE protocol layer/sub-system. "PDM" "PDCP" "RLC" "MAC" "L1A" "RRC" "NAS" "TIMERS" "LOG" "CRITICALERR" "MEM" "L1AEXT" "ALL" "PWR" "USIM" "MBMS" "PWR_BREAKDOWN" "COMPRESSION" "RPM" "ALL" "QNORBLK"
<filter>	String type. Used to reduce the command output to the information defined by the filter: "TX" "RX"
<counter_type>	String type. Used to reduce the command output to specific type defined by: "IPBYTES"
<action>	String type. Action on counter values: "CLEAR" Erase all counter values to zero
<stats_string>	String defined in an arbitrary format for specified layer counters starting with textual prefix - "LAYER Stats:".
<errcode>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFGEXT="stats","PDM",,"CLEAR" //Clears the counters.
OK
```


2.1.2. Platform-Related AT Commands

2.1.2.1. AT+QCFGEXT="quecopen"* Enable/Disable QuecOpen Function

This command enables/disables your application to load the QuecOpen function or queries the current setting.

AT+QCFGEXT="quecopen" Enable/Disable QuecOpen Function	
Write Command AT+QCFGEXT="quecopen"[,<value>]]	Response If the optional parameter is omitted, query the current setting: +QCFGEXT: "quecopen",<value> OK If the parameter is specified, set whether to enable QuecOpen function: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<value>	Integer type. Enable/disable the QuecOpen function. <u>0</u> Enable 1 Disable
----------------------	---

2.1.2.2. AT+QCFGEXT="ver" Query SW/FW Version Information

This command queries SW/FW version information.

AT+QCFGEXT="ver" Query SW/FW Version Information	
Write Command AT+QCFGEXT="ver"	Response <ver_info> OK

	If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<ver_info>	String type. Version information.
<errcode>	Integer type. Error code. See Chapter 3 for more information.

Example

```

AT+QCFGEXT="VER"
NP Package: RK_03_00_00_00_32
Apps: RKAPP_03_00_00_00_32_f719e1efa50b6c3a1091fb448534210e31546a42
Using APP processor - no SB or 3B versions
MAC Revision: REL_1250_03_00_00_REV_220389
MAC Package Version: ALT1250_03_00_00_00_32_FW
MAC Build Time: Dec_04_2019_17_34_09
PHY Revision: 12.50.220357
PHY Build Time: Dec_04_2019_15_17_29
PHY Build Info: releas_0
PMP Revision: 220389
PMP Version: 32_FW
PMP build time: Dec_04_2019_17_34_09
DSP Revision: 57939
BB Product: 1250
BB HW Revision: 20
RFIC_6200 Revision: 00
NP Build Time: Dec 4 2019 18:57:16
C&V Ver: 9.27
Calibration date: 30.12.18
Calibrated NP package: RK_02_01_00_00_68

OK

```

2.1.2.3. AT+QCFGEXT="debug" Configure Debug Mode

This command enables/disables real-time or offline debugging or queries the current setting.

AT+QCFGEXT="debug" Configure Debug Mode	
Write Command AT+QCFGEXT="debug"[,<value>]	Response If the optional parameter is omitted, query the current setting: +QCFGEXT: "debug",<value> OK If the optional parameter is specified, configure debug mode: OK If there is any error: ERROR
Maximum Response Time	4500 ms
Characteristics	This command takes effect after rebooting. The configuration is saved automatically.

Parameter

<value>	Integer type. Debug mode settings.
0	Disable
1	Enable (real-time, DBG_UART, AUX_UART)
2	Log dump (manual, DBG_UART)
3	Log dump (manual, <i>ufs:/logs</i>)
4	Unhide auto log dumps (<i>ufs:/logs</i>)

NOTE

If debug mode is enabled, entering PSM mode might not be possible.

2.1.2.4. AT+QCFGEXT="devinfo" Query Device Identification Values

This command queries device identification and device characterization values from different types of persistent memories.

AT+QCFGEXT="devinfo" Get Device Identification Values

Write Command AT+QCFGEXT="devinfo"	Response <type1>,<value1> [<type2>,<value2>] ... [<typeN>,<valueN>] OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<type>	String type.	
"DeviceSerialNumber"	Returned string: board serial number.	
"ModelNumber"	Returned string: vendor model ID number.	
"SiliconPartNumber"	Returned string: silicon part number.	
"SiliconID"	Returned hexadecimal 16-byte individual silicon identifier.	
"OtpVersion"	Returned integer value of OTP version.	
"TestProgramRev"	Returned integer value of test program revision.	
"SiliconLotId"	Returned string: physical location of the silicon die: lot ID.	
"SiliconWaferId"	Returned integer value: physical location of the silicon die: wafer ID.	
"SiliconXLoc"	Returned integer value: physical location of the silicon die: X location.	
"SiliconYLoc"	Returned integer value: physical location of the silicon die: Y location.	
"MinActiveVoltage"	Returned integer value of minimal voltage for digital core operation (VDD min in active mode) in mV.	
"MinRetentionVoltage"	Returned integer value of minimal voltage for retention mode (logic and memory) in mV.	
"Mculsolation"	Reported boolean value for MCU from Modem security isolation and protection, "true" if enabled.	

	"RomSecureBoot"	Reported boolean value for ROM Secure Boot feature, "true" if enabled.
	"ModemSecureBoot"	Reported boolean value for Modem Secure Boot feature, "true" if enabled.
	"McuSecureBoot"	Reported boolean value for MCU Secure Boot feature, "true" if enabled.
	"FlashPartNumber"	Returned string: flash part number.
<value>	Integer without quotes or string/hex parameter in quotes.	
<errcode>	Integer type. Error code. See Chapter 3 for more information.	

2.1.2.5. AT+QCFGEXT="meas" Return the Measurement for Specified Measurement Type

This command returns the measurement for the specified measurement type.

AT+QCFGEXT="meas" Return the Measurement for Specified Measurement Type	
Write Command AT+QCFGEXT="meas",<measurement_type>	Response For RSRP, SINR: <measurement_type>: Reported=<measurement_value>,Rx0Tx0=<measurement_value>,Rx0Tx1=<measurement_value>,Rx0Tx2=<measurement_value>,Rx0Tx3=<measurement_value>,Rx1Tx0=<measurement_value>,Rx1Tx1=<measurement_value>,Rx1Tx2=<measurement_value>,Rx1Tx3=<measurement_value> For RSRQ, RSSI: <measurement_type>:Reported=<measurement_value>, Rx0Tx0=<measurement_value>,Rx0Tx1=<measurement_value>, Rx1Tx0=<measurement_value>,Rx1Tx1=<measurement_value> For TX Power: TX Power: PUSCH=<measurement_value>, PUCCH=<measurement_value>, PRACH=<measurement_value>, SRS=<measurement_value> For Temperature, Pathloss: <measurement_type>: <measurement_value> For CQI: CQI: <measurement_value>

For Signal Quality:

Signal Quality:RSRP=<measurement_value>,RSRQ=<measurement_value>,SINR=<measurement_value>,RSSI=<measurement_value>

For RS_SNR:

RS_SNR=<measurement_value>

For RS_SINR:

RS_SINR=<measurement_value>

For Power Headroom:

PHR=<measurement_value>, PHR Level=<measurement_value>

For Srxlev:

SRXLEV=<measurement_value>

For Network Time corresponding to SFN of serving cell (AT+QCFGEXT="meas","93"):

NWTIME:<networkTTI>,<networkUtcTime>

For E-CID (AT+QCFGEXT="meas","95"):

ECID:<gcid>[,<TimeDifIndex>,<ta>,<MCC>,<MNC>,<TAC>,<EARFCN>,<cellID>,<SFN>,<RSRP>,<RSRQ>[,<EARFCN>,<cellID>,<SFN>,<RSRP>,<RSRQ>[...]]]

For NBS RSRP in compressed format:

NBS RSRP:<EARFCN>,<cellID>,<measurement_value> [,<EARFCN>,<cellID>,<measurement_value>[...]]

For all neighboring NBS simultaneous RSRP and RSRQ reporting:

EARFCN=<EARFCN>,CellID=<cellID>,RSRP=<measurement_value>,RSRQ=<measurement_value> [EARFCN=<EARFCN>,CellID=<cellID>,RSRP=<measurement_value>,RSRQ=<measurement_value>] [...]

For all neighboring cells RSRP, RSRQ and RSSI:

EARFCN=<EARFCN>,CellID=<cellID>,<measurement_type>=<measurement_value> [EARFCN=<EARFCN>,CellID=<cellID>,<measurement_t

	type>=<measurement_value> [...] OK If there is an error related to ME functionality: +CME ERROR: <errcode> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<measurement_type>	String type. Measurement type that should be a number in quotes when used in the command, whereas it should be a string without quotes in the response. "0" RSRP "1" RSRQ "2" SINR "3" RSSI "4" TX Power - Not applicable for NB-IoT in RRC IDLE mode "5" Temperature "6" Pathloss "7" CQI (Channel Quality Indicator) - Not applicable for NB-IoT "8" Signal Quality (RSRP & RSRQ & SINR & RSSI) "10" RSRP reported value only "11" RSRQ reported value only "12" SINR reported value only "13" RS_SNR (reference signal signal-to-noise ratio) "14" RS_SINR (reference signal signal-to-interference-plus-noise ratio) "15" Power Headroom - Only supported for NB-IoT "16" Srxlev as defined in 36.304 "17"-"92" Reserved "93" Network Time alignment with SFN "95" Measurements for E-CID "96" RSRP for all detected NBS (same as "98") in compressed format: in single line and each piece of eNB measurement data (<EARFCN>,<cellID>,<measurement_value>) is separated by an additional space.
---------------------------------	---

	"97" RSRP & RSRQ for all detected NBS
	"98" RSRP for all detected NBS
	"99" RSRQ for all detected NBS
	"100" RSSI for all detected NBS
<measurement_value>	Integer type. dBm for RSRP (<RSRP>), RSSI dB for RSRQ (<RSRQ>), SINR, Pathloss 0.1 dBm for TX Power (for example, 2.5 dBm = 25) Degrees (°C) for Temperature 0.1 dB for RS_SINR, RS_SINR, PHR (for example, 2.5 dB = 25) Measurement range: -140 ≤ RSRP ≤ 0 dBm -60 ≤ RSRQ ≤ 0 dB -128 ≤ SINR ≤ 40 dB -26 ≤ TX Power ≤ 40 -128 ≤ Temperature ≤ 128 0 ≤ CQI ≤ 15 -12.0 ≤ RS_SINR/RS_SINR ≤ 40.0 If RSRP/RSRQ measurement value for an antenna is not supported, the command returns "N/S" (not supported indication for this specific antenna) in the returned string.
<networkTTI>	Integer type. Subframe counter of the serving cell corresponding to network UTC time. It is a decimal running from 0 to 10239 (i.e., rollover at 10240) also known as TTI (Transmission Time Interval) counter.
<networkUtcTime>	Integer type. This field specifies the network UTC time corresponding to the specified TTI counter. UTC time is a decimal counter of 1 msec unit counted since 00:00:00 on 1 January, 1900.
<gcid>	Hex type. Global cell ID hexadecimal value.
<TimeDifIndex>	Integer type. RxTxTimeDiff decimal index (as defined in 9.1.9.2 of 3GPP 36.133) of the measured cell. The value should be reported by MAC based on RxTxTimeDiff reported by PHY. Note that RxTxTimeDiff used by the PHY is different from the value received by MAC CE and has better Ts granularity and accuracy.
<ta>	Integer type. Currently-used Timing Advance value (N_{TA}) of the measured cell. The N_{TA} value is represented by index values of $TA = 0, 1, 2, \dots, 1282$, where an amount of the time alignment is given by $N_{TA} = TA \times 16$ per 3GPP 36.213.
<MCC>	String type. A three-digit value indicating mobile country code as defined in ITU-T Recommendation E.212 Annex A.
<MNC>	String type. A three-digit or two-digit value indicating the mobile network code as defined in ITU-T Recommendation E.212 Annex A.
<TAC>	Hex type. Two-byte tracking area code in hexadecimal format.

<EARFCN>	Integer type. Decimal EARFCN value.
<cellID>	Integer type. Decimal Physical Cell ID value.
<SFN>	Integer type. The decimal system frame number (SFN) of the measured cell during which measurement has been performed. Since there is averaging over multiple SFN, it is advised to supply the latest SFN. If a value is not available at the time of the query, command returns N/A (without quotes).
<errcode>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFGEXT="meas","0"
```

```
RSRP: Reported = -90, Rx0Tx0 = -92, Rx0Tx1 = -88, Rx0Tx2 = -140, Rx0Tx3 = -140, Rx1Tx0 = -140, Rx1Tx1 = -140, Rx1Tx2 = -140, Rx1Tx3 = -140
```

```
OK
```

NOTE

1. SINR is not reported over the air. Its reported value contains combined measurement values for all antennas.
2. The reported range for **<measurement_value>** parameter is wider than the range defined for Measurement Reporting in 3GPP specification. It is intended to report weak and abnormal measurements, such as jamming detection.

2.1.2.6. AT+QCFGEXT="setbdelay" Set U-Boot Delay

This command modifies the U-Boot delay applied during the next cold boot or queries the current setting. Once modified, the new U-Boot delay will be used as the new default timeout in all following cold boots.

AT+QCFGEXT="setbdelay" Set U-Boot Delay

Write Command

```
AT+QCFGEXT="setbdelay"[,<mode>]
```

Response

If the optional parameter is omitted, query the current setting:
+QCFGEXT: "setbdelay",<mode>

OK

If the optional parameter is specified, set U-Boot delay:

OK

If there is any error:

ERROR

Maximum Response Time	300 ms
Characteristics	The command takes effect after rebooting. The configuration is saved automatically.

Parameter

<mode>	Integer type. U-Boot delay.
0	0 second
1	3 seconds

2.1.2.7. AT+QCFGEXT="copstout" Set Maximum Timeout for AT+COPS=? Command

The command queries or sets **AT+COPS=?** timeout. **AT+COPS=?** scans all set bands. Scanning time depends on the number of set bands. If timeout is less than scanning time, **AT+COPS=?** will return an empty list. In that case, timeout must be adjusted with **AT+QCFGEXT="copstout"**.

AT+QCFGEXT="copstout" Set Maximum Timeout for AT+COPS=? Command	
Write Command AT+QCFGEXT="copstout"[,<timeout>]	Response If the optional parameter is omitted, query the current setting: +QCFGEXT: "copstout",<timeout> OK If the optional parameter is specified, set timeout: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<timeout>	Integer type. Range: 210 – 600 seconds. Default timeout: 210 seconds.
------------------------	--

2.1.3. GNSS-Related AT Commands

2.1.3.1. AT+QCFGEXT="addgeo" Add a Geo-fence

This command adds a geo-fence or queries the current setting.

AT+QCFGEXT="addgeo" Add a Geo-fence	
Write Command	Response
AT+QCFGEXT="addgeo",<geoid>[,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]]]	If all parameters after "addgeo" are omitted, query the current setting of all geo-fences that have been added: [+QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] ... +QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] OK If the parameters after <geoid> are omitted, query the current setting of the specified geo-fence: +QCFGEXT: "addgeo",<geoid>,<mode>,<shape>,<lat1>,<lon1>,<lat2>,<lon2>,<lat3>,<lon3>,<lat4>,<lon4>]]] OK If <shape>=0, add a circular geo-fence and the parameters after <lat2> must be omitted. OK If <shape>=1, add a circular geo-fence and the parameters after <lon2> must be omitted. OK If <shape>=2, add a triangular geo-fence and the parameters after <lon3> must be omitted. OK If <shape>=3, add a quadrangle geo-fence and all parameters must be specified. OK If there is any error related to ME functionality: +CME ERROR: <errcode>

Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–9.
<mode>	Integer type. URC report mode. 0 Disable the URC reported when entering or exiting the geo-fence 1 Enable the URC reported when entering the geo-fence 2 Enable the URC reported when exiting the geo-fence 3 Enable the URC reported when entering or exiting the geo-fence See Chapter 2.2.1 for more information about the URC.
<shape>	Integer type. Geo-fence shape. 0 Circularity with center and radius 1 Circularity with center and one point on the circle 2 Triangle 3 Quadrangle
<lat1>	The latitude of the point defined as the center of the geo-fence circular region or the first point. Unit: degree. Format: $\pm dd.dxxxxx$. Range: -90.000000 to 90.000000. (- South latitude)
<lon1>	The longitude of the point defined as the center of the geo-fence circular region or the first point. Unit: degree. Format: $\pm ddd.dxxxxx$. Range: -180.000000 to 180.000000. (- West longitude)
<lat2>	When <shape> is 0, this parameter is a radius. Range: 0–6000000. Unit: meter. When <shape> is not 0, this parameter is a latitude of the point on the circle or the second point. Unit: degree. Format: $\pm dd.dxxxxx$. Range: -90.000000 to 90.000000. If <shape> is 0, the parameters after <lat2> must be omitted.
<lon2>	The longitude of the point on the circle or the second point. Unit: degree. Format: $\pm ddd.dxxxxx$. Range: -180.000000 to 180.000000. If <shape> is 1, the parameters after <lon2> must be omitted.
<lat3>	The latitude of the third point. Unit: degree. Format: $\pm dd.dxxxxx$. Range: -90.000000 to 90.000000.
<lon3>	The longitude of the third point. Unit: degree. Format: $\pm ddd.dxxxxx$. Range: -180.000000 to 180.000000. If <shape> is 2, the parameters after <lon3> must be omitted.
<lat4>	The latitude of the fourth point. Unit: degree. Format: $\pm dd.dxxxxx$. Range: -90.000000 to 90.000000.
<lon4>	The longitude of the fourth point. Unit: degree. Format: $\pm ddd.dxxxxx$. Range: -180.000000 to 180.000000.
<errcode>	Integer type. Error code. See Chapter 3 for more information.

2.1.3.2. AT+QCFGEXT="deletegeo" Delete a Geo-fence

This command deletes a geo-fence.

AT+QCFGEXT="deletegeo" Delete a Geo-fence	
Write Command AT+QCFGEXT="deletegeo",<geoid>	Response OK If there is any error related to ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–10. If 10, delete all geo-fences.
<errcode>	Integer type. Error code. See Chapter 3 for more information.

NOTE

On BG951A-GL module supports only up to 5 **<geoid>s**. ID 5 means deleting all geo-fences.

2.1.3.3. AT+QCFGEXT="querygeo" Query the Position Relative to Geo-fence

This command queries the position relative to the geo-fence.

AT+QCFGEXT="querygeo" Query the Position Relative to Geo-fence	
Write Command AT+QCFGEXT="querygeo",<geoid>	Response +QCFGEXT: "querygeo",<geoid>,<pos_wrt_geo-fence> OK If there is any error related to ME functionality: +CME ERROR: <errcode>
Maximum Response Time	300 ms
Characteristics	/

Parameter

<geoid>	Integer type. Geo-fence ID. Range: 0–9.
<pos_wrt_geofence>	Integer type. Position relative to geo-fence. 0 Position unknown 1 Position inside the geo-fence 2 Position outside the geo-fence
<errcode>	Integer type. Result code. See Chapter 3 for more information.

2.1.3.4. AT+QCFGEXT="xonoff" Enable Software Flow Control

This command enables software flow control.

AT+QCFGEXT="xonoff " Enable Software Flow Control	
Write Command AT+QCFGEXT="xonoff"[,<sw_flow_en>]	Response +QCFGEXT: "xonoff",<sw_flow_en> OK If the optional parameter is specified, set software flow control: OK
Maximum Response Time	300 ms
Characteristics	/

Parameter

<sw_flow_en>	Integer type. 0 Disable Software flow control 1 Enable Software flow control
--------------	--

2.2. Description of URCs

2.2.1. +QIND: "GEOFENCE" Indicate Entering or Leaving Geo-fence

+QIND: "GEOFENCE" Indicate Entering or Leaving Geo-fence	
+QIND: "GEOFENCE",<geoid>,<act>	The URC indicates entering or leaving a geo-fence.

```
ion>,<time>,<latitude>,<longitude>,<altitude>,<course>,<speed>,<PDOP>,<HDOP>,<VDOP>
```

Parameter

<geoid>	Integer type. ID of the entered or left geo-fence.
<action>	Integer type. Current action of the module. 1 Entering the geo-fence 2 Exiting the geo-fence
<time>	UTC time when entering or leaving the geo-fence. Format: YY/MM/DD hh:mm:ss
<latitude>	Latitude of the module when entering or leaving the geo-fence. Unit: degree. Format: ±dd.dddddd. Range: -90.000000 to 90.000000.
<longitude>	Longitude of the module when entering or leaving the geo-fence. Unit: degree. Format: ±ddd.dddddd. Range: -180.000000 to 180.000000.
<altitude>	Float type. Mean sea level altitude. Unit: meter.
<course>	Float type. Course over ground, relative to true north. Unit: degree.
<speed>	Float type. Speed over ground. Unit: m/s
<PDOP>	Float type. Position dilution of precision.
<HDOP>	Float type. Horizontal dilution of precision.
<VDOP>	Float type. Vertical dilution of precision.

2.2.2. +QIND: "nipd","recv" Indicate Incoming Non-IP Data

After receiving the non-IP data from the MT, the module reports the URC **+QIND: "nipd","recv"** to notify the host about incoming data. Then the host can retrieve data via **AT+QCFGEXT="nipdr"**. If the buffer is not empty, and the module receives data again, it does not report a new URC until all the received data have been retrieved from the buffer via **AT+QCFGEXT="nipdr"**. The buffer size is 2048 bytes. If the received data exceed the buffer size, subsequent data will be discarded.

+QIND: "nipd","recv" Indicate Incoming Non-IP Data

+QIND: "nipd","recv"	The URC notifies the host about incoming data from the network. Then the host can retrieve the data via AT+QCFGEXT="nipdr" .
----------------------	---

2.2.3. +QIND: "nipd","close" Indicate the Connection is Closed

+QIND: "nipd","close" Indicate Connection is Closed

+QIND: "nipd","close"

The URC notifies the user that the NIDD connection has been closed accidentally. If the connection closed normally via **AT+QCFGEXT="nipd",0**, this URC is not reported.

3 Summary of <errcode>

The error code **<errcode>** indicates a final result or an error related to mobile equipment or network operation. The definition of **<errcode>** is provided in the table below.

Table 3: Summary of <errcode>

<errcode>	Definition
0	Operation successful
501	Invalid parameter
517	Geo-fence ID does not exist
651	Invalid parameter for non-IP data
652	Non-IP data sending error
653	Network error
654	NIDD operation in process
655	Timeout error
656	NIDD connection not opened
657	NIDD connection already opened
658	NIDD connection accidentally closed

4 Appendix References

Table 4: Related Document

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual

Table 5: Terms and Abbreviations

Abbreviation	Description
APN	Access Point Name
APP	Application
ASCII	American Standard Code for Information Interchange
CE	Control Element
CQI	Channel Quality Indicator
DFOTA	Delta Firmware Upgrade Over-The-Air
DSP	Digital Signal Processor
EARFCN	E-UTRA Absolute Radio Frequency Channel Number
E-CID	Enhanced Cell ID
eNB	Evolved Node B
FW	Firmware
HDOP	Horizontal Dilution of Precision
ID	Identifier
IP	Internet Protocol

IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
ITU-T	International Telecommunication Union Telecommunication Standardization Sector
LTE	Long-Term Evolution
MAC	Medium Access Control
MCC	Mobile Country Code
MCU	Microcontroller Unit
ME	Mobile Equipment
MNC	Mobile Network Code
MO	Mobile Originated
MT	Mobile Terminated/Mobile Termination
NAS	Non-Access Stratum
NB-IoT	Narrowband Internet of Things
NIDD	Non-IP Data Delivery
N/S	Not Supported
N _{TA}	Timing Advance Value
OTP	One Time Programmable
PDCP	Packet Data Convergence Protocol
PDOP	Position Dilution of Precision
PDM	Pulse Density Modulation
PHR	Power Headroom Report
PHY	Physical
PMP	Power Management Processor
RFIC	Radio Frequency Integrated Circuit
RLC	Radio Link Control

ROM	Read Only Memory
RRC	Radio Resource Control
RSRP	Received Signal Code Power
RSRQ	Reference Signal Received Quality
RSSI	Received Signal Strength Indicator
RX	Receive
SFN	System Frame Number
SINR	Signal-to-Interference-plus-Noise Ratio
SNR	Signal-to-Noise Ratio
SW	Software
TTI	Transmission Time Interval
TX	Transmit
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment
URC	Unsolicited Result Code
USIM	Universal Subscriber Identity Module
UTC	Coordinated Universal Time
VDD	Drain Voltage
VDOP	Vertical Dilution of Precision
